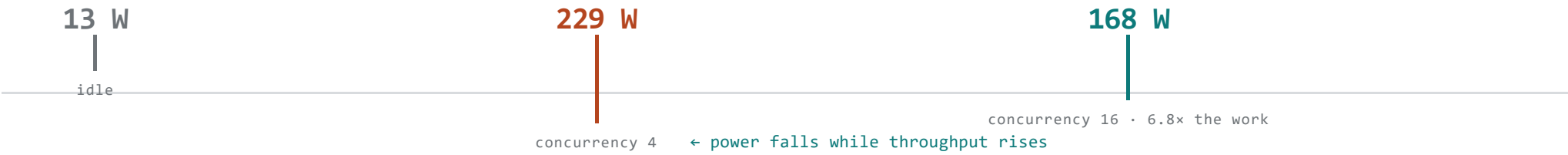


costbench

On-prem AI agents have no invoice. costbench gives them one — priced in **joules measured on the card that ran the work**.

AMD Radeon PRO W7900 (gfx1100) · ROCm 7.2.1 · llama.cpp HIP · Qwen2.5-Coder 7B & 14B Q4_K_M · egress: on-prem only

GPU BOARD POWER — MEASURED, SYSFS POWER1_AVERAGE



696 tok/s

sustained on one W7900
7B, concurrency 16

8.7_x

energy efficiency from batching
8.70× on 7B · 8.80× on 14B

−88%

energy cost per token
paid with 7.8× time to first token

The gain from batching does not depend on the model

Two models, one twice the size of the other. Normalise each curve by its own concurrency-1 value and they coincide within 1%.

tokens per joule, 7B vs 14B

FINDING 1

Relative efficiency gain is model-independent

8.70× on the 7B, 8.80× on the 14B — a property of the architecture, not of one model. Four independent sessions, two load patterns, achieved concurrency verified per run.

FINDING 2

Absolute efficiency scales inversely with model size

The 14B/7B tokens-per-joule ratio is 0.53 / 0.52 / 0.54 at concurrency 1 / 4 / 16. Double the weights, half the efficiency: the signature of a memory-bandwidth-bound decode.

WHAT IT BUYS

A bill for on-prem inference

A gateway meters every request against the measured curve; an agent reports what each task cost. One recorded task: 194 tokens · 404 J · €0.000028 · 0 bytes egress — the zero verified against a local audit ledger, not asserted.

FOUR MEASUREMENT BUGS FOUND AND FIXED — EACH PRODUCED PLAUSIBLE-LOOKING DATA

Sensor lag ~18 s

power1_average is a long-window mean; 2–5 s runs reported 14 W under full load. Fixed with 90 s runs, first 25 s discarded.

Shared 8-GPU host

sysfs exposes every card; the first readable path was somebody else's. GPU now resolved by PCI bus, recorded in every result.

Wave-barrier load

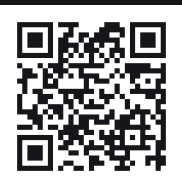
waiting for all N leaves an idle tail that grows with N, deflating power at high batch. Replaced by closed-loop load.

Stale server on the port

a "14B" run re-measured the 7B. The served model is now verified, and the invalid run is kept in rejected/.



REPOSITORY



DEMO 4:00

github.com/Artkill24/costbench

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Every figure on this poster is **measured** and reproducible from **proofs/** — raw results, logs, audit exports, SHA256SUMS. Declared limits: GPU board power only; the cause of the power drop is hypothesised, not proven.